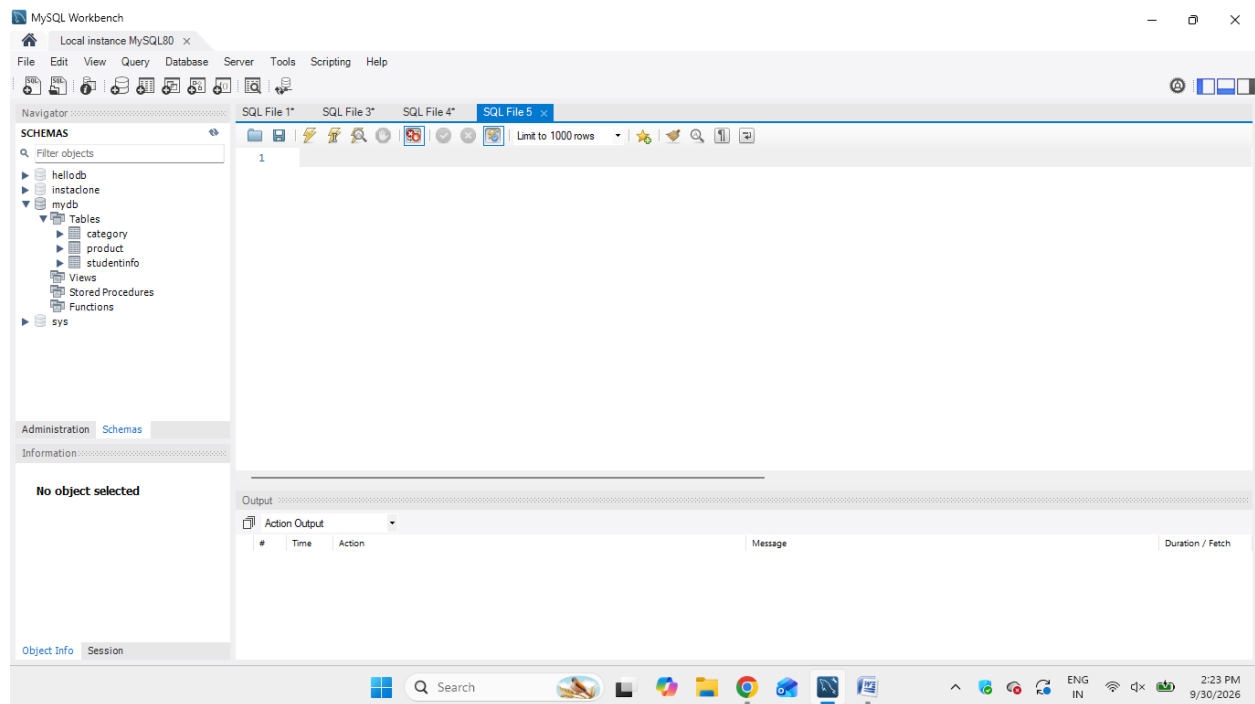


Session 01 Introduction to Databases
Trainer Coverage: What is a database, DBMS, RDBMS (with examples). Difference between MySQL, PostgreSQL, Oracle, SQLite. Tables, rows, columns, and relationships. Demo: Show data stored in MySQL Workbench.

Q:-01 Install MySQL Workbench on your computer, connect to the local MySQL server, and take a screenshot of the default databases shown in the Schemas panel.

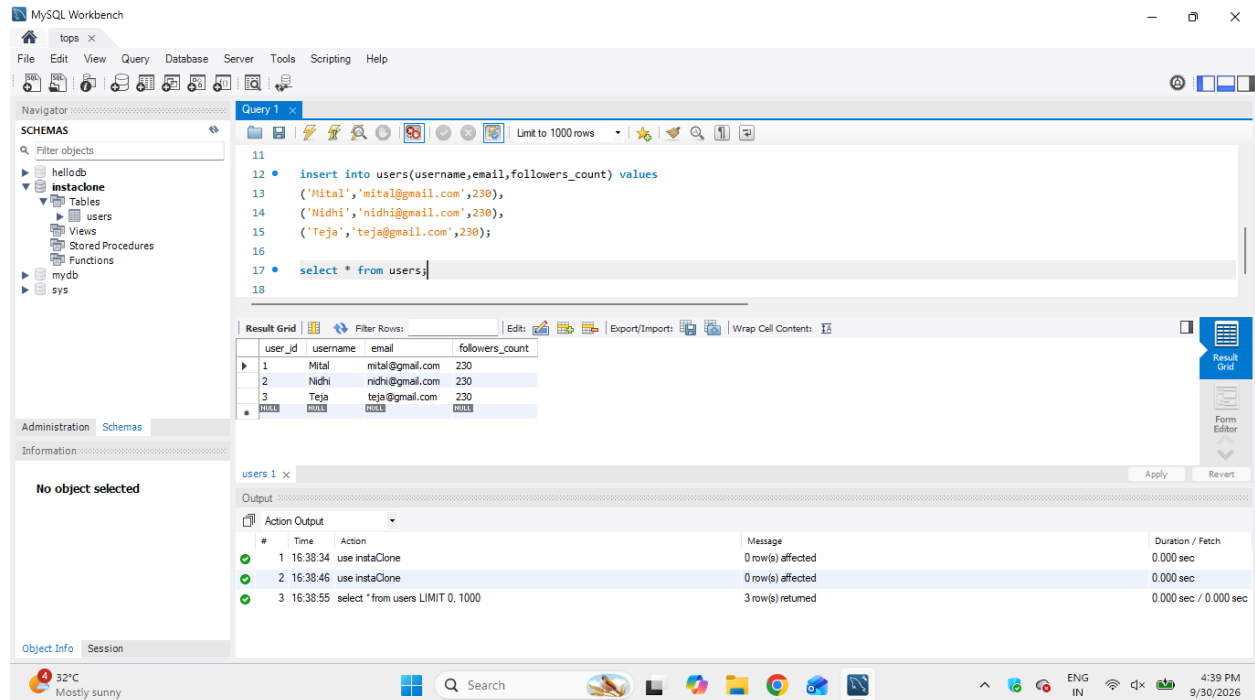


Q:-02 Create a new database called InstaClone in MySQL Workbench, then create a table named Users with columns: user_id (INT, primary key), username (VARCHAR), email (VARCHAR), and followers_count (INT).

```
create table users(  
  user_id integer primary key auto_increment,  
  username VARCHAR(100),  
  email VARCHAR(100),  
  followers_count integer);
```

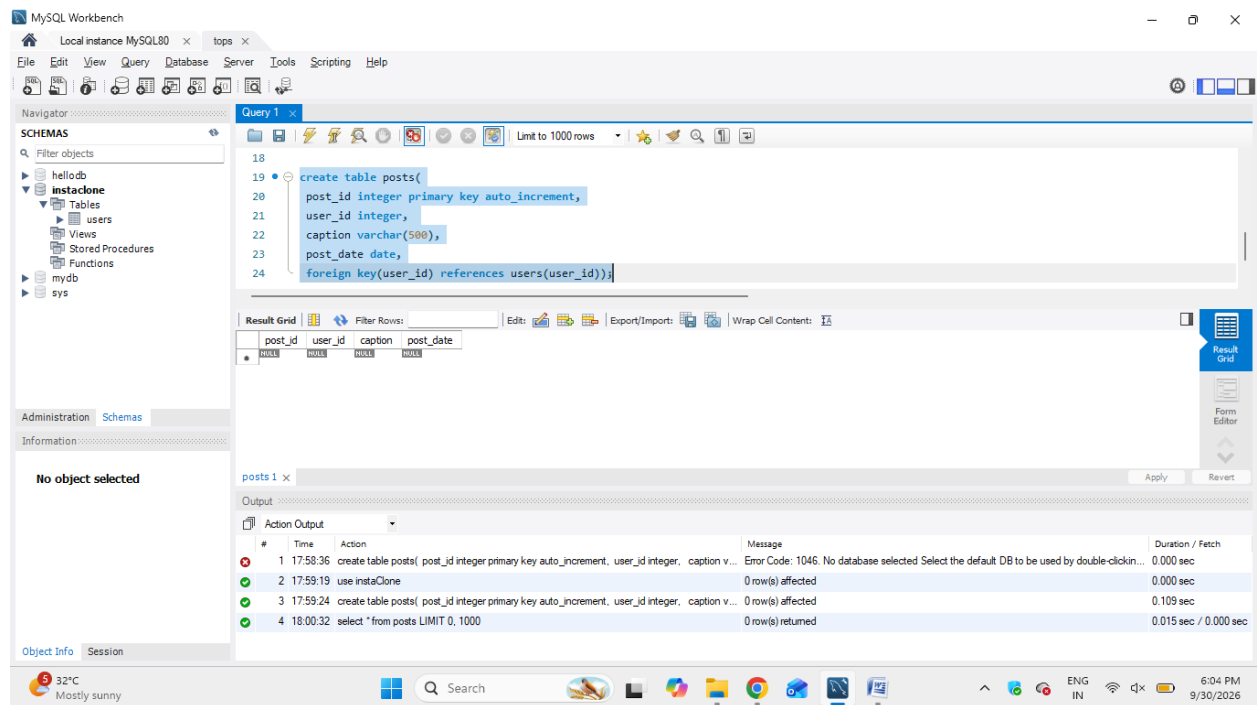
Q:-03 Insert 3 sample users into the Users table you created, using realistic Instagram-style usernames and follower counts.

```
insert into users(username,email,followers_count) values  
( 'Mital','mital@gmail.com',230),  
( 'Nidhi','nidhi@gmail.com',230),  
( 'Teja','teja@gmail.com',230);
```



Q:-04 Create another table in the InstaClone database called Posts with columns: post_id (INT, primary key), user_id (INT), caption (VARCHAR), and post_date (DATE). Add a foreign key from Posts.user_id to Users.user_id to establish a relationship.

```
create table posts(  
  post_id integer primary key auto_increment,  
  user_id integer,  
  caption varchar(500),  
  post_date date,  
  foreign key(user_id) references users(user_id));
```



Q:-05 Write a short comparison (3-4 lines each) of MySQL, PostgreSQL, Oracle, and SQLite, focusing on where each one is commonly used (for example: web apps, mobile apps, enterprise systems, etc.).

1. MySQL

- Commonly used for **web applications** and websites.
- Popular with PHP, Java, Python, and Node.js applications.
- Used for e-commerce, content management systems, and business applications.
- Easy to learn and widely supported.

2. PostgreSQL

- Commonly used for **modern web applications and data-intensive systems**.
- Strong support for complex queries and advanced data types.
- Used in analytics, SaaS applications, and large-scale applications.

- Known for reliability and standards compliance.

3. Oracle Database

- Commonly used in **large enterprise systems**.
- Used by banks, insurance companies, governments, and large organizations.
- Designed for high-volume transactions, security, and complex business systems.
- Often used where advanced enterprise database features are required.

4. SQLite

- Commonly used in **mobile apps, desktop applications, and embedded systems**.
- The database is stored directly in a single local file.
- Widely used in Android/iOS apps for local data storage.
- No separate database server is required.